AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 22 · STUDENT LESSON

Requests Responses and JSON

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

Week 8: JSON validity is separate from value correctness.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Parse JSON into Python records. Try the recall before opening notes.

Your learning target

I can parse **JSON** into Python records.

I can validate data after **parsing**.

I can document **input** and **output contract**s.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Tools and APIs. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from User Need and AI Contribution. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 22: **Request**s, **Response**s, and **JSON**

Clear **interface**s specify fields, types, errors, and ownership.



CREATOR MISSION

Validate sample **request/response** objects.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> Why is reading text different from validating it?

This week's end goal

By the end, I can explain **request**s **response**s and **json** using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Requests Responses and JSON case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **request**s, **response**s, and **json** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A fictional activity service supplies **JSON** text. The **input contract** is a **JSON object** with name text and minutes an integer from 0 through 60. The **output contract** will be a record with accepted and minutes fields.

Parsing valid **JSON** is only the first step.

STUDENT LESSON · 3 OF 18

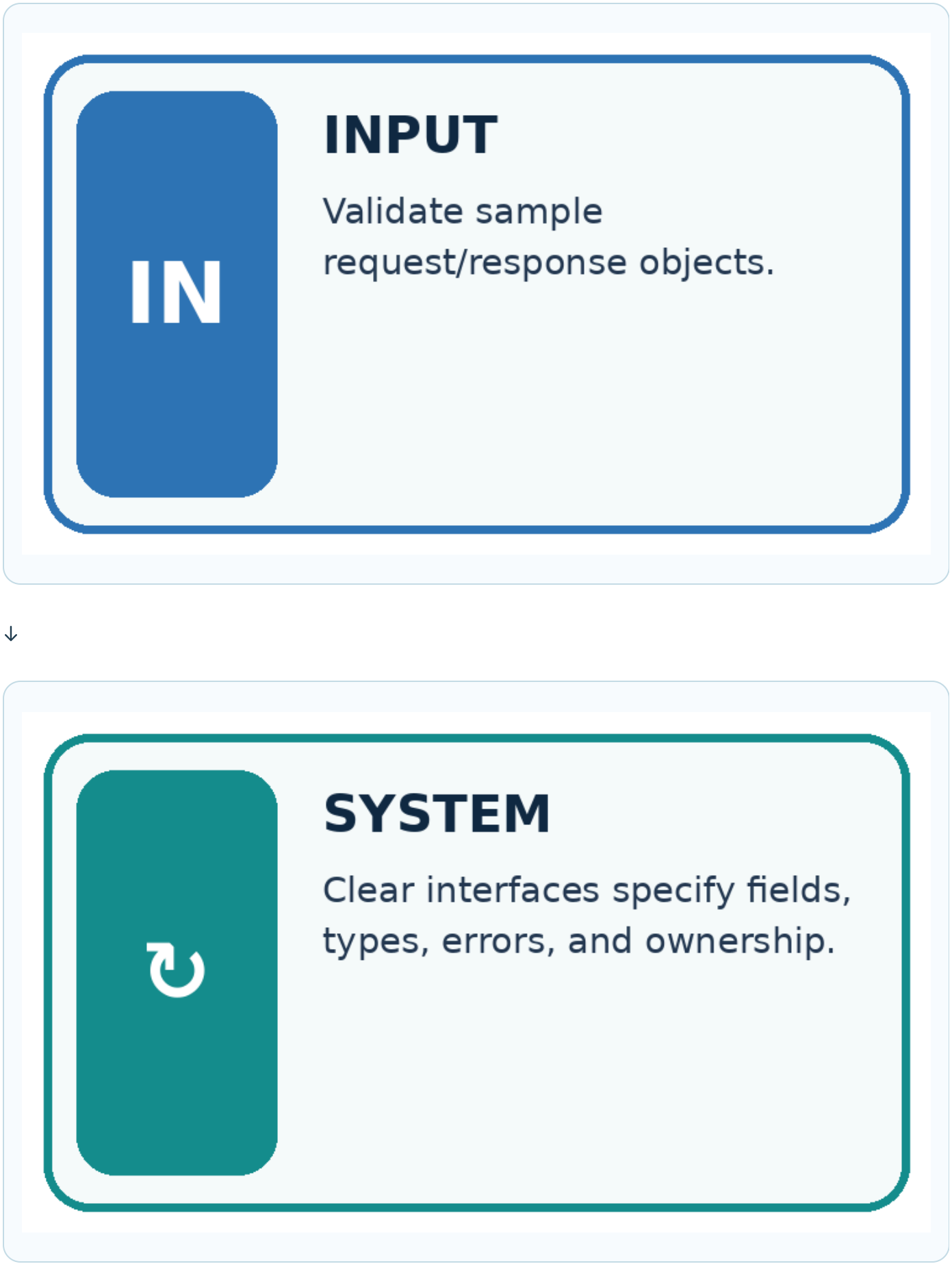
See how it works

> Follow the mechanism for Requests Responses and JSON. Connect each labelled input, deciding step and result.

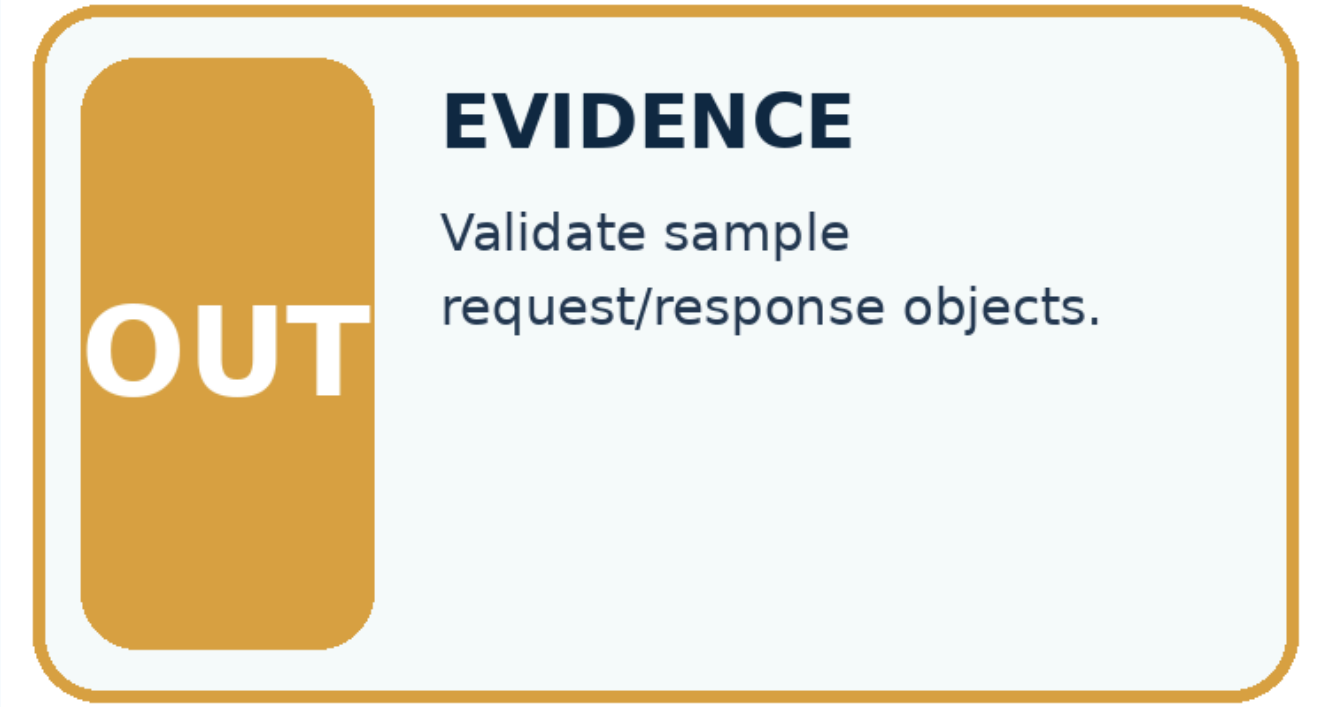
VISUAL MECHANISM

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Validate sample request/response objects.
SYSTEM
Clear interfaces specify fields, types, errors, and ownership.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

```
import json
text = '{"name": "Sketch", "minutes": 12}'
record = json.loads(text)
valid = (type(record) is dict
         and type(record.get("name")) is str
         and len(record["name"]) > 0
         and type(record.get("minutes")) is int
         and 0 <= record["minutes"] <= 60)
```

```
result = {"accepted": valid, "minutes": record.get("minutes")}
print(json.dumps(result))
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Input text	JSON object	Parse with json.loads
Python value	Dictionary fields	Check schema
Output text	accepted and minutes	Serialize with json.dumps

Read each row from left to right and connect the evidence to the stated rule.

The **output** is {"accepted": true, "minutes": 12}. **JSON** uses lowercase true, false and null; Python uses True, False and None. **json**.loads converts between representations.

The worked **output** line assumes the parsed top-level value is a dictionary; a robust **interface** must reject other top-level types before accessing fields.

Check the envelope before its fields

JSON is a text format for exchanging values. **Parsing** asks whether the text follows that format. Schema checking asks whether the parsed value has the structure and values your task requires.

Start by checking that the whole value is an object, represented here by a Python dictionary. Only then look up name and minutes. A valid **JSON** list is still the wrong top-level structure for this **contract**.

Syntax success and acceptance are separate decisions.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

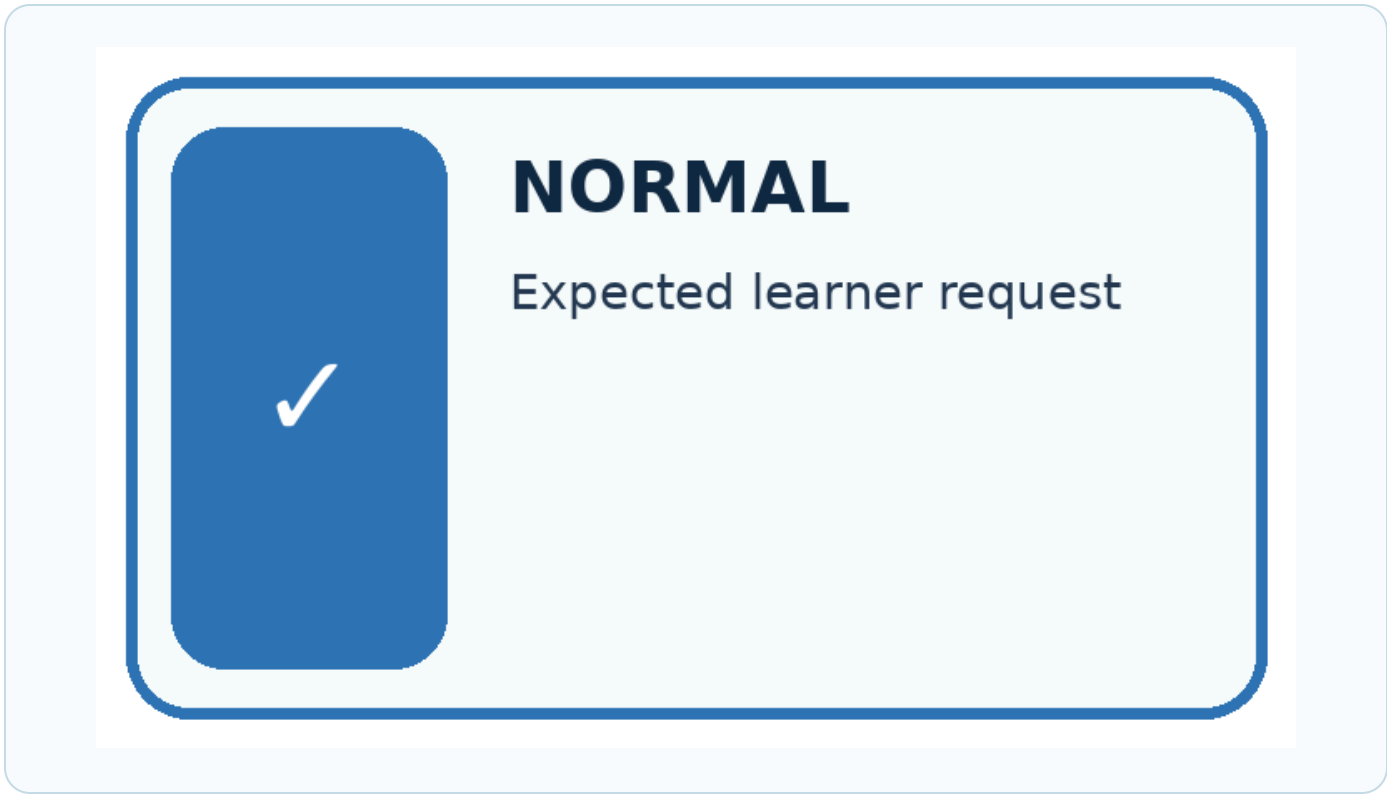
> Compare the supplied cases for Requests Responses and JSON. Identify the detail that changes the conclusion.

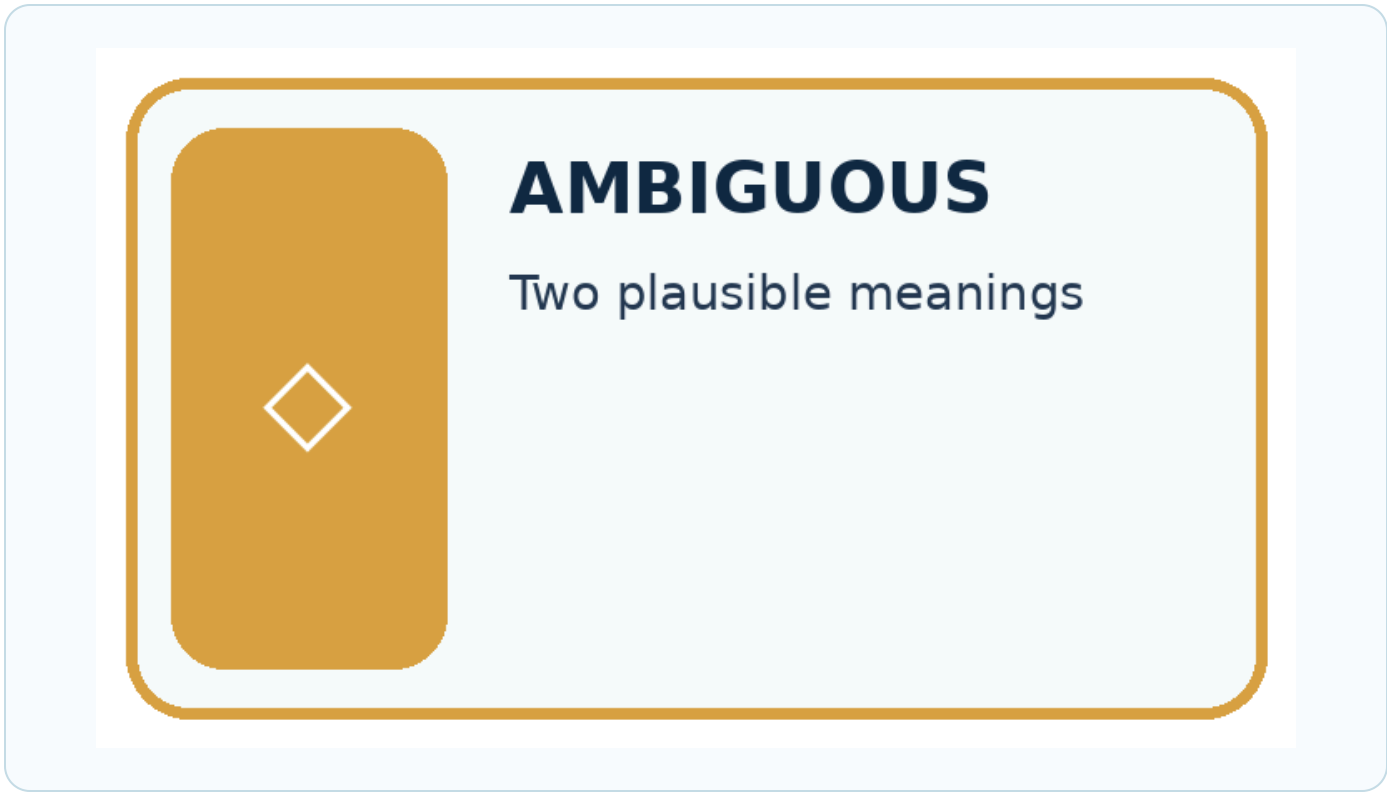
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.


NORMAL
Expected learner request


AMBIGUOUS
Two plausible meanings

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Replace 12 with the text "12".

Predicted result

Prediction: Reject the **response** under this strict **contract**.

Why the result follows

Reason: Digits inside quotes represent text, whereas portions requires an integer.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

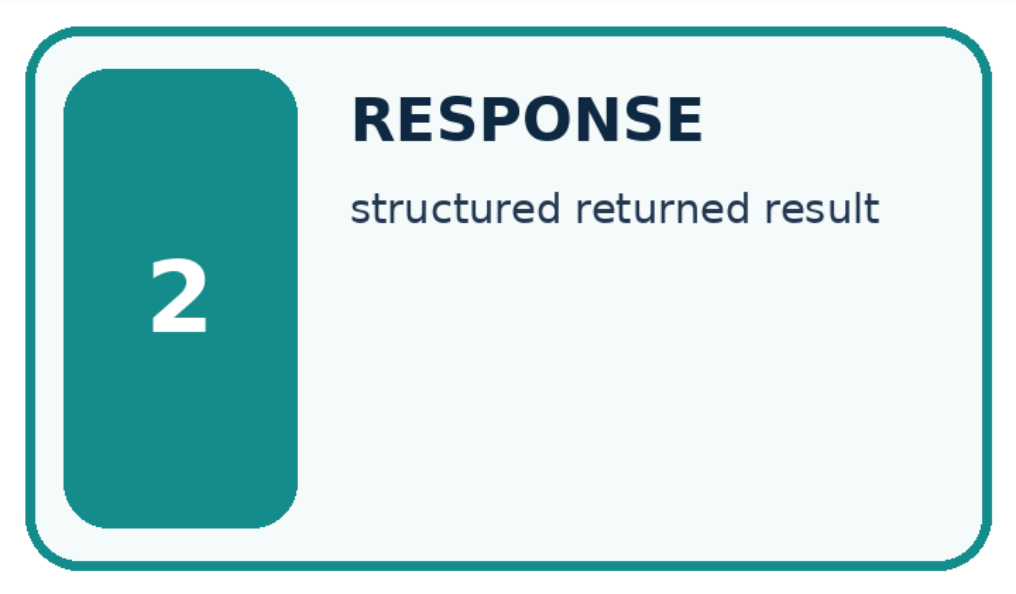
> Use the vocabulary of Requests Responses and JSON to explain a concrete decision, rather than only memorising definitions.

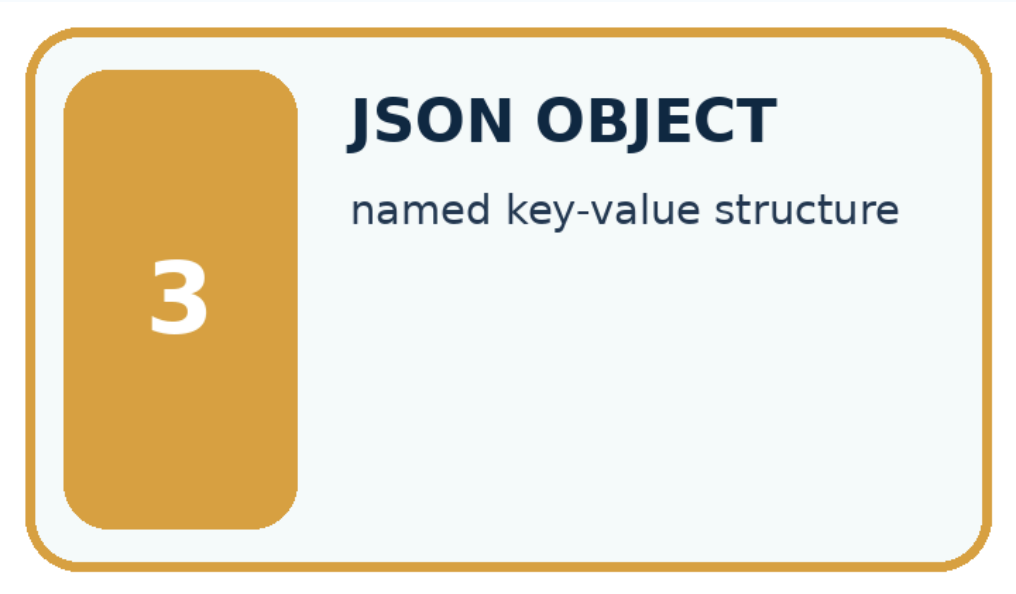
VOCABULARY LAB

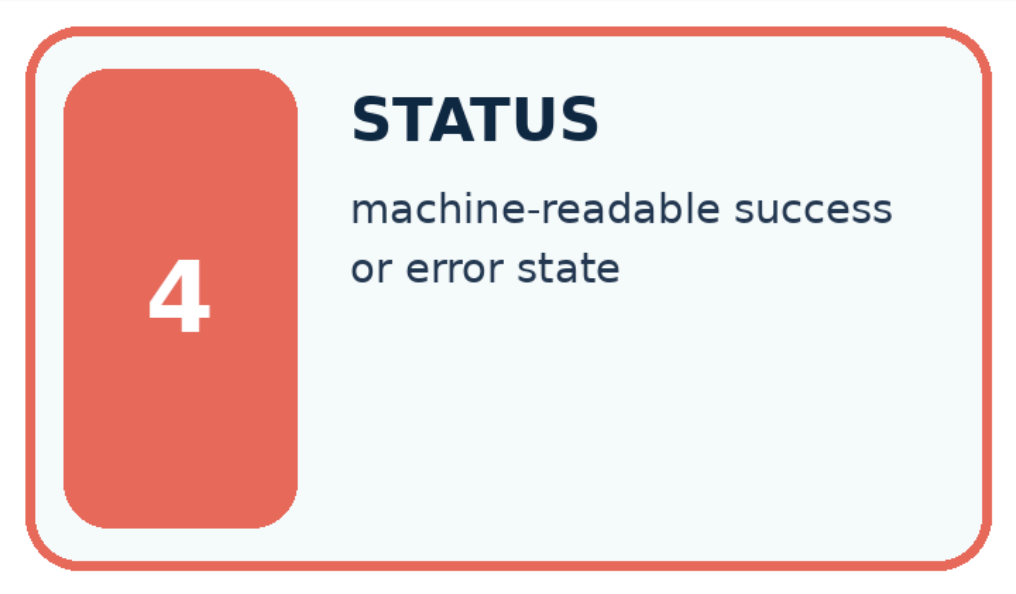
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
REQUEST
structured message asking for an operation


2
RESPONSE
structured returned result





Word	My own example	Not this / boundary
request	A message asking a system to return information or perform an action.	
response	The information or status returned after a request.	
JSON object	A collection of named fields and values written using JSON syntax, such as {"pages":80}.	

Word	My own example	Not this / boundary
status	A reported condition of a request or process, such as success, pending or error.	

Words used in the closer explanation

Word	Meaning
JSON	A text format for structured values
Parsing	Converting formatted text into program values
Interface	The agreed way one part of a system exchanges data with another
Contract	Specified inputs, outputs and error behavior

STUDENT LESSON · 6 OF 18

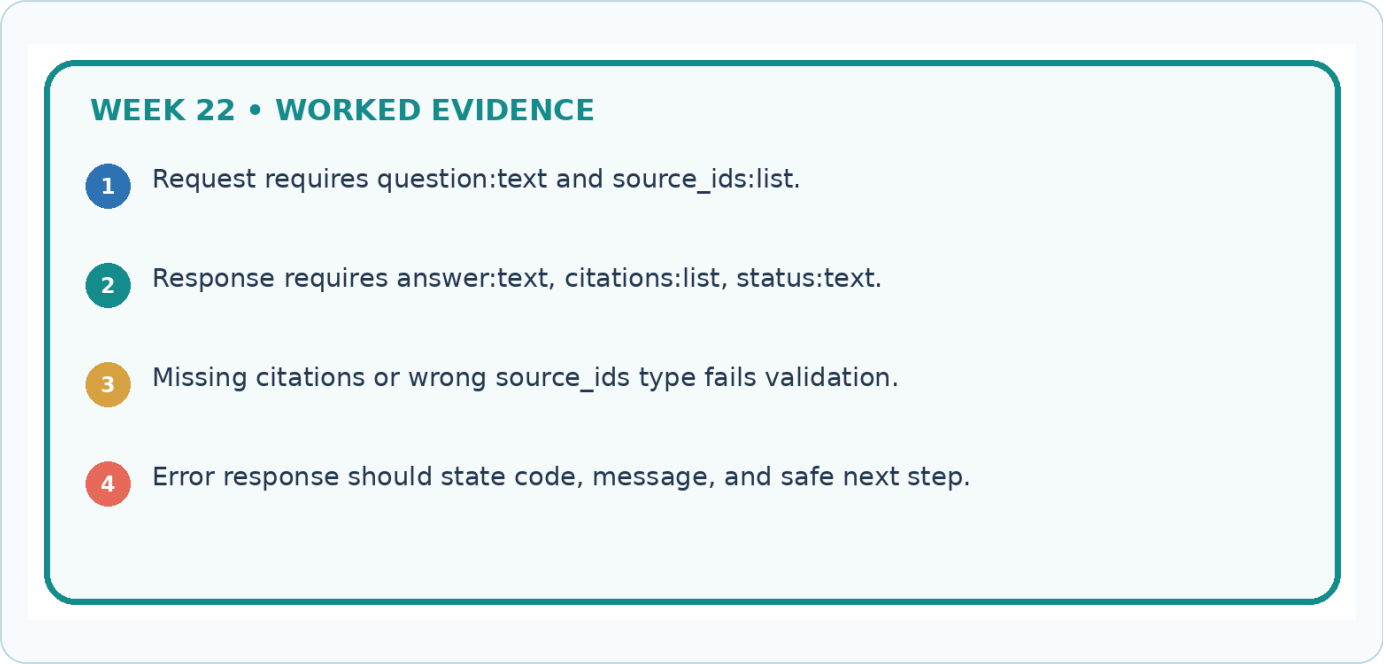
Follow a worked example

> Follow the worked trace for Requests Responses and JSON, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. **Request** requires question:text and source_ids:list.

2. **Response** requires answer:text, citations:list, **status**:text.

3. Missing citations or wrong source_ids type fails validation.

4. Error **response** should state code, message, and safe next step.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

JSON text contains {"name": "Sketch", "minutes": "12"}. Does parsing fail, or does validation fail?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Requests Responses and JSON

A lunch-count **response** must satisfy a declared **JSON contract**.

The facts we will use

Supplied facts

The toy **response** has exactly class_code text and portions integer 0..30. Example: {"class_code":"R8","portions": 12}.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/22

Requests Responses and JSON

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Read fields

class_code is nonempty text; portions is the integer 12.

↓

2. Check range

0 <= 12 <= 30, so the declared count is within range.

↓

3. Accept structure

Accept this response shape; acceptance does not prove that 12 portions were actually counted.

CHANGE ONE THING

Replace 12 with the text "12".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read fields	class_code is nonempty text; portions is the integer 12.
2. Check range	0 <= 12 <= 30, so the declared count is within range.
3. Accept structure	Accept this response shape; acceptance does not prove that 12 portions were actually counted.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

JSON well-formedness, schema validity and real-world truth are different checks.

Keep the stated limit

Read the labelled picture: Would converting the text silently prove the source count is correct? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Replace 12 with the text "12". Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Requests Responses and JSON claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Valid JSON syntax proves the values are meaningful and safe. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

An **interface** should state what happens for malformed text, wrong structure and invalid values. Do not expose raw private **input** in error messages. Use only the fictional records supplied for practice.

A JSON record parses successfully. Does that prove its observation is correct?

STUDENT LESSON · 8 OF 18

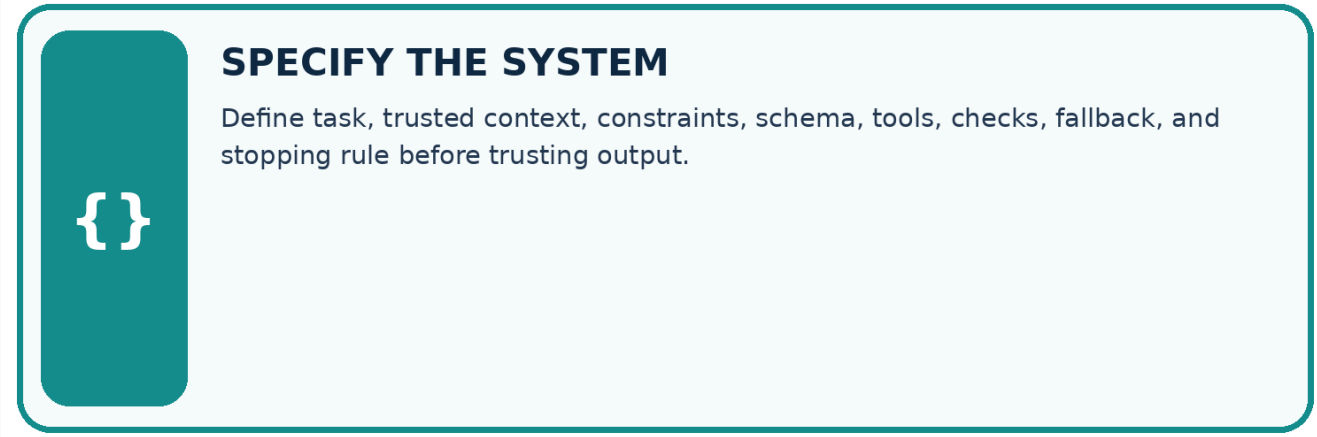
Practise together

Practise Requests Responses and JSON with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Single quotes around property names are not valid **JSON** syntax. Use double quotes in the **JSON** text. If the whole parsed value is a list, do not call .get on it: check the top-level type first and report a schema failure.

Practise with less help: Separate failures

Use workbook W2 for the connected example "**JSON** and **interface contract**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **JSON** and **interface contract**s. Proposed explanation: "A successful run proves the conclusion". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Would converting the text silently prove the source count is correct? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Requests Responses and JSON. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Inspect eight objects for syntax, required fields, types, allowed values, and source ownership; repair without inventing data.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

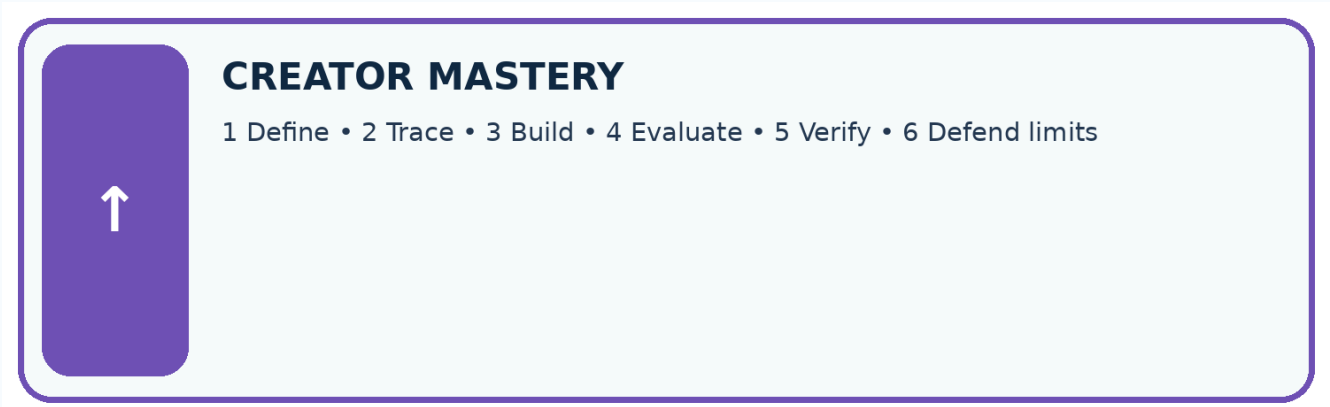
Build your understanding

Move from recall to tracing, diagnosis and transfer in Requests Responses and JSON. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

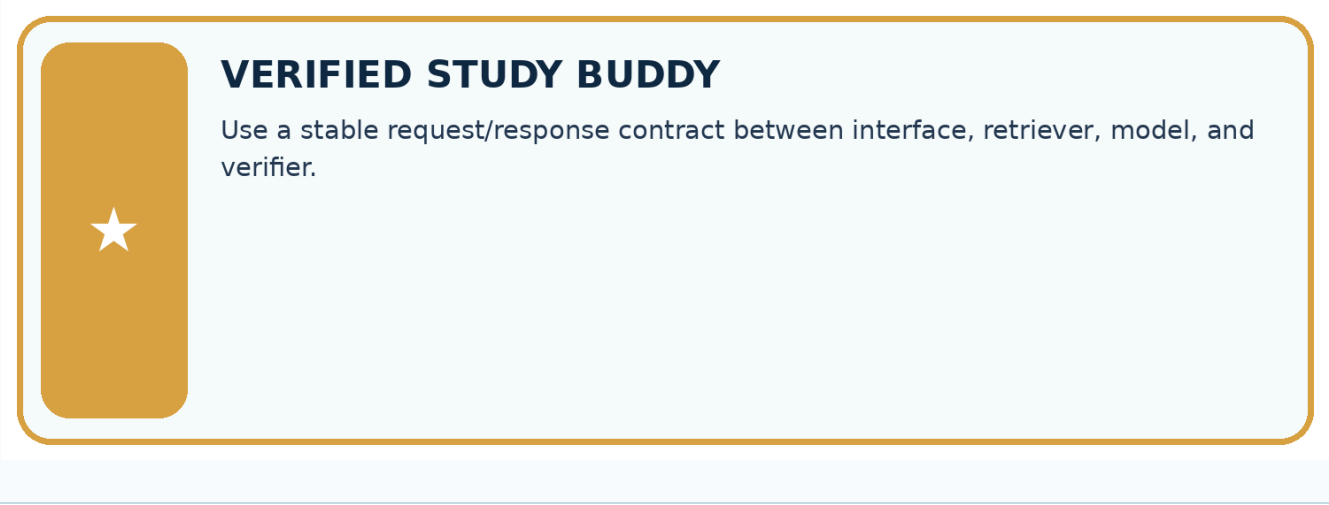
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Use a stable **request/response contract** between **interface**, retriever, model, and verifier.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Validate sample **request/response** objects.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

Study a complete case

> Revisit the complete Requests Responses and JSON case for supported practice. Repeated examples do not become fresh assessments.

Week 22 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Request is for B7; **response** is {"id":"B7","pages":120}. Another **response** has malformed **JSON**.

The question

> What checks precede use?

Worked explanation

Request success, **JSON** validity, matching id, required fields and numeric type. Handle malformed **response**s explicitly.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Validate the API response before using it

Evidence or stage	Value or result
Request success	Handle failure explicitly
JSON syntax	Reject malformed data
Identity	id must match B7
Field type	pages must be numeric 120

Request is for B7; **response** is {"id":"B7","pages":120}. Another **response** has malformed **JSON**.

> What checks precede use?

Request success, **JSON** validity, matching id, required fields and numeric type. Handle malformed **response**s explicitly.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Specify independently

Define a **JSON contract** for {"label": "Box", "count": 4}: nonempty text label, integer count 0 through 10. Give a valid case, a wrong-type case and a missing-field case with expected acceptance.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 22 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

An API returns {"ok":false,"error":"not_found"}. Your app tries to display **response**.title. What should it do instead?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Use a stable **request/response contract** between **interface**, retriever, model, and verifier.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Request id=7 returns **status**=200 and {"id":8,"pages":90}. Explain why a successful **status** is insufficient and specify the handling.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Requests Responses and JSON, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A lunch-count **response** must satisfy a declared **JSON contract**.

By the end, I can	How I will show it
Parse JSON into Python records	Explain the deciding step, show a trace or test, and state one limit.
Validate data after parsing	Explain the deciding step, show a trace or test, and state one limit.
Document input and output contracts	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Requests Responses and JSON without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Parse **JSON** into Python records

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A lunch-count **response** must satisfy a declared **JSON contract**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Accept this **response** shape; acceptance does not prove that 12 portions were actually counted.

1. Accept structure

2. Read fields

3. Check range

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Accept structure Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. Read the picture in order.

Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace 12 with the text "12". Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Reject the **response** under this strict **contract**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reject the **response** under this

strict **contract**. Digits inside quotes represent text, whereas portions requires an integer.

Repair a missing building block

Review Tools and APIs (ai-3/21). Goal: Explain the **mechanism** and terminology of tools and api **contract**s. Short practice: Does a valid C7 ID permit the extra lending action?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review User Need and AI Contribution (ai-3/19). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What non-AI baseline would reveal whether this helper adds value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Verify Numbers and Dates (ai-3/16). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Which part can be corrected immediately from existing **evidence**, and which needs follow-up?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

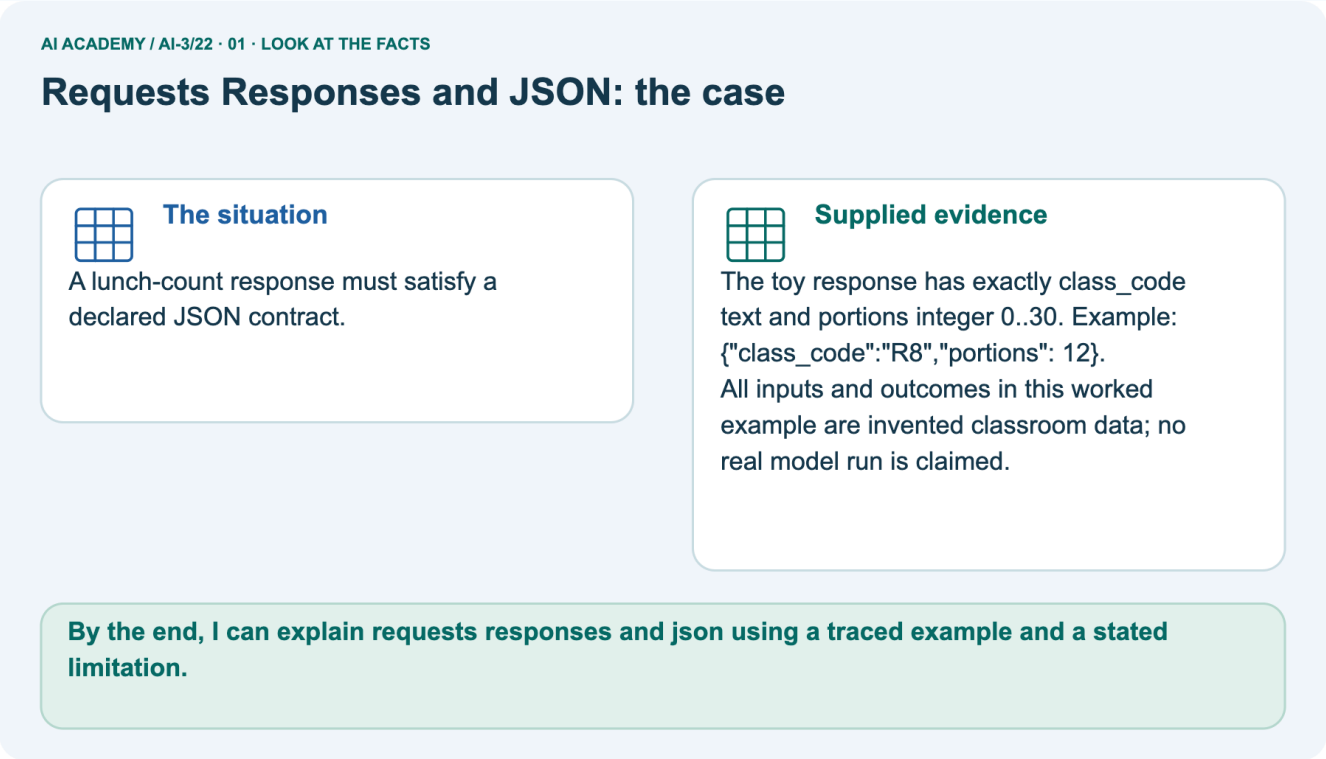
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Requests Responses and JSON graphic using its labels, deciding evidence and relevant limitation.

01. Requests Responses and JSON: the case



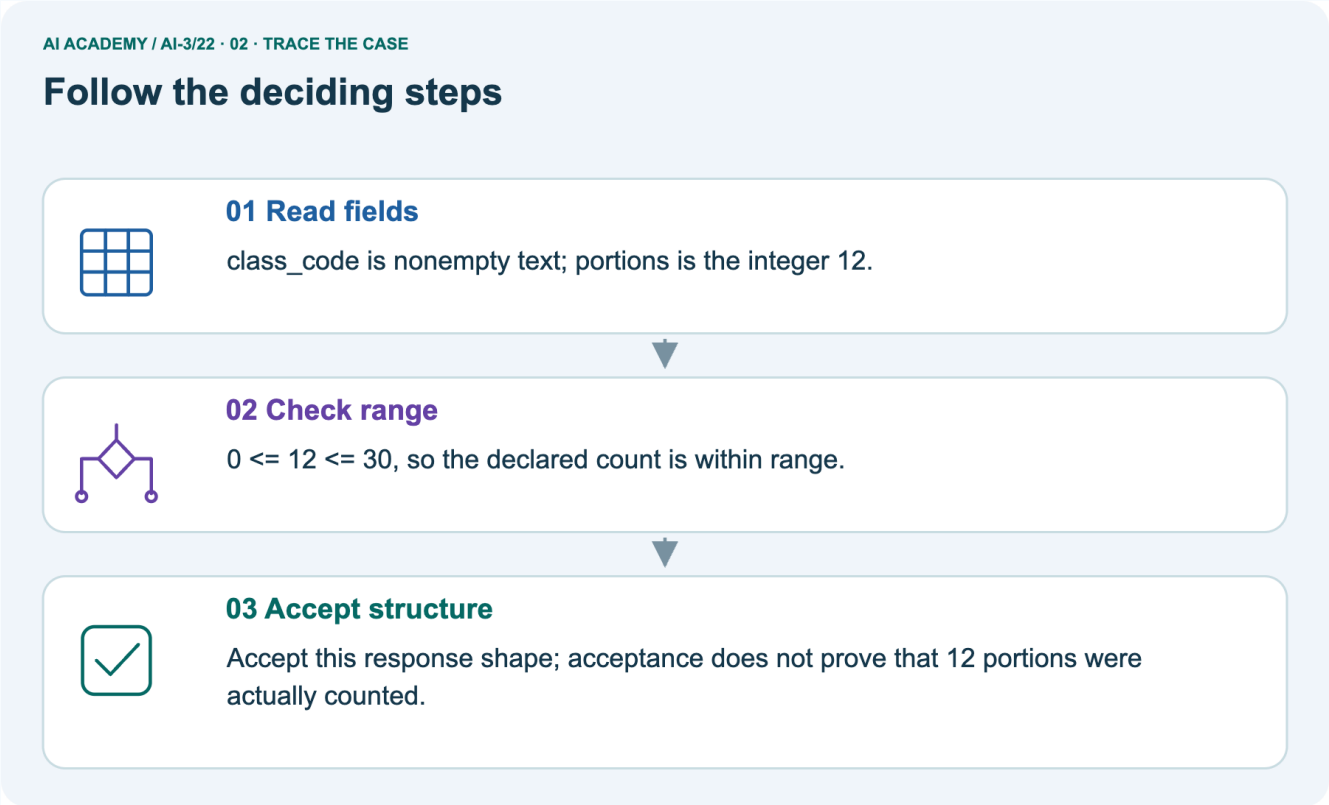
Follow the labelled picture with the matching explanation. Open the vector diagram

A lunch-count **response** must satisfy a declared **JSON contract**. Supplied facts: The toy **response** has exactly class_code text and portions integer 0..30. Example: {"class_code":"R8","portions": 12}.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

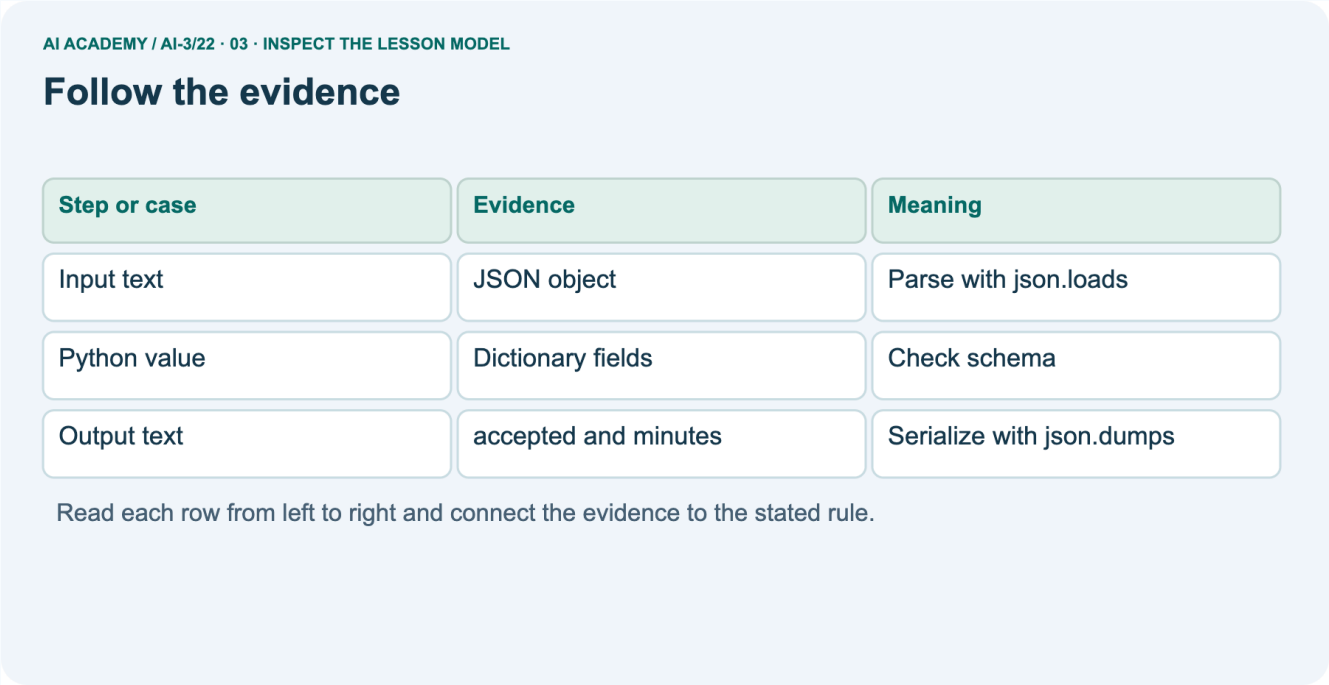


Follow the labelled picture with the matching explanation. Open the vector diagram

Read fields: class_code is nonempty text; portions is the integer 12. → Check range: 0 <= 12 <= 30, so the declared count is within range. → Accept structure: Accept this **response** shape; acceptance does not prove that 12 portions were actually counted.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

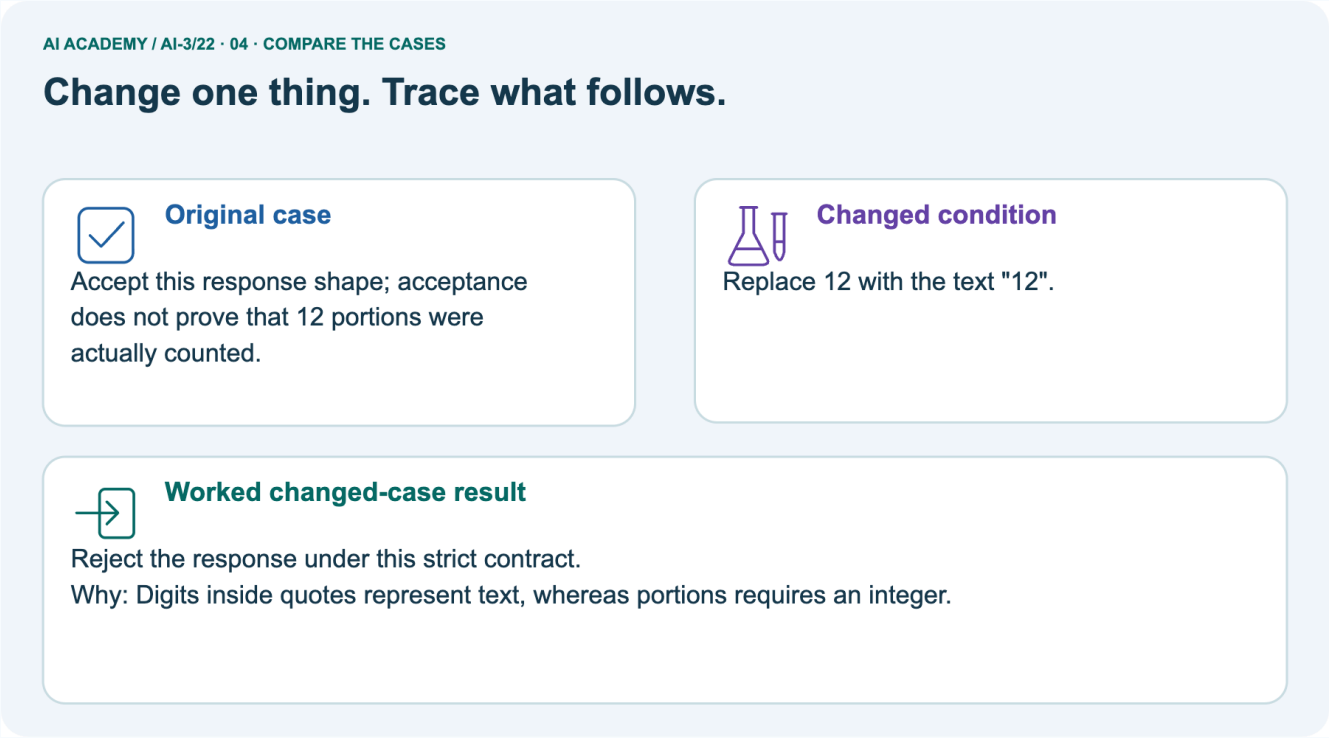


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. **Input** text | **JSON object** | Parse with **json**.loads. Python value | Dictionary fields | Check schema. **Output** text | accepted and minutes | Serialize with **json**.dumps

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



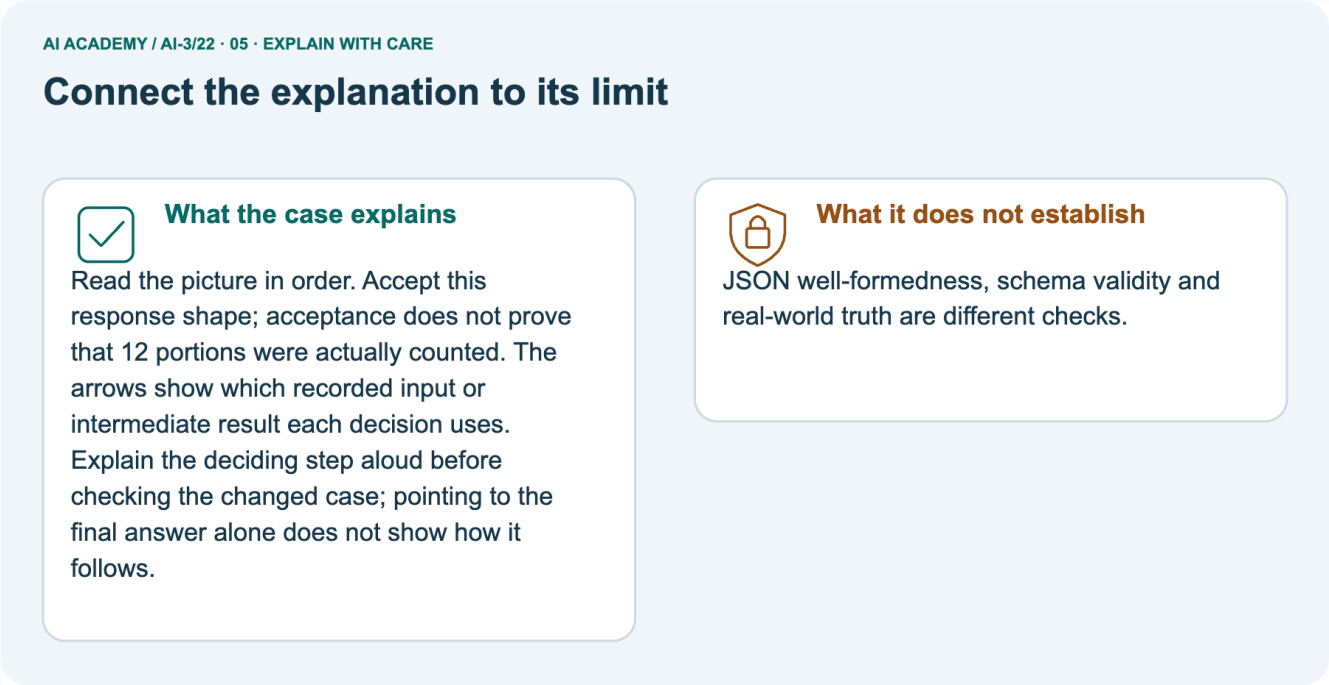
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. Changed condition: Replace 12 with the text "12". Expected result: Reject the **response** under this strict **contract**.

Reason: Digits inside quotes represent text, whereas portions requires an integer.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



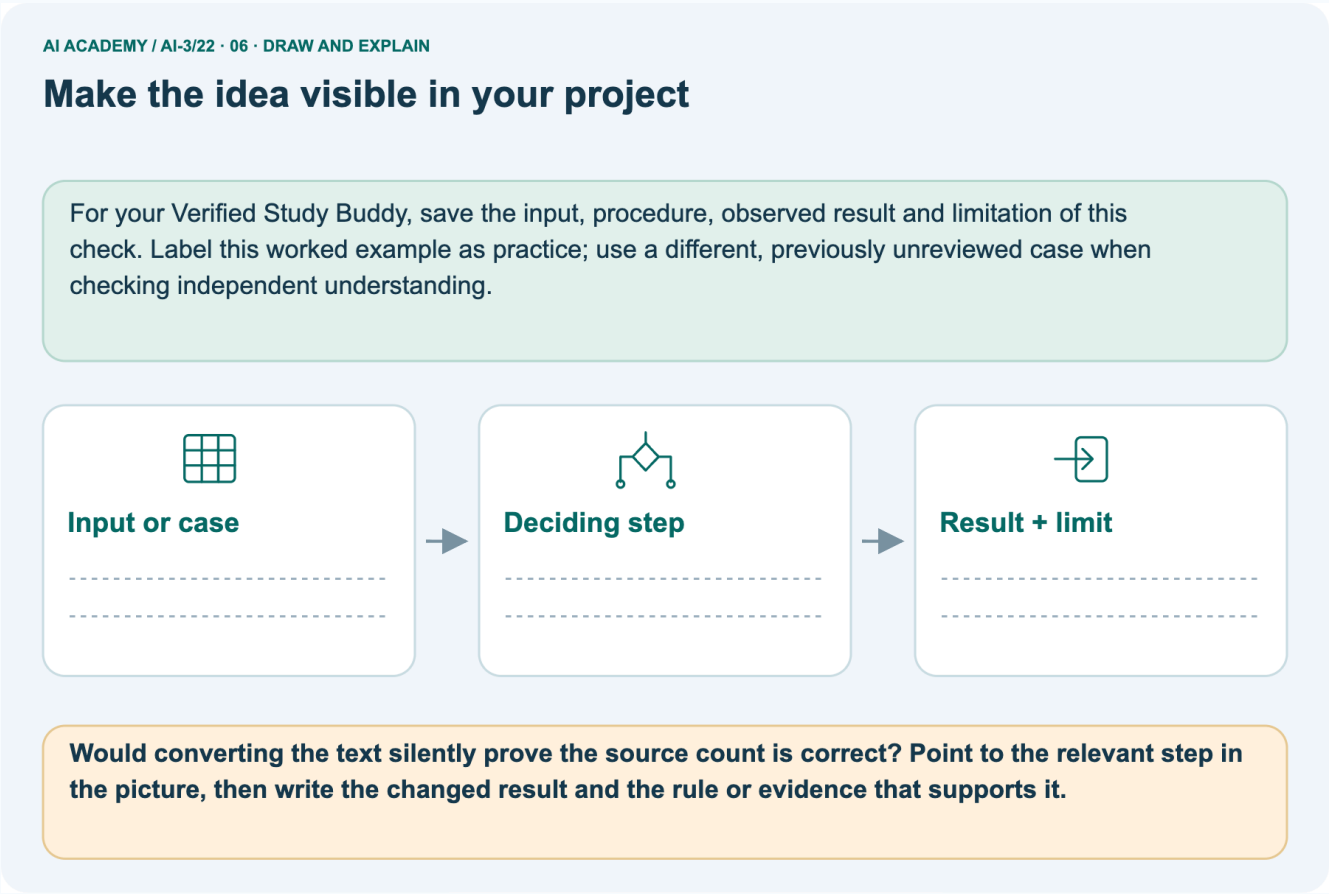
Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: **JSON** well-formedness, schema validity and real-world truth are different checks.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Would converting the text silently prove the source count is correct? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.